A 3 nm FinFET 125 TOPS/W-29 TFLOPS/W, 90 TOPS/mm^2-17 TFLOPS/mm^2 SRAM-Based INT8, and FP16 Digital-CIM Compiler With Support for Multi-Weight Update/Cycle

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***Abstract*—This article presents an static random-access memory (SRAM)-based digital compute-in-memory (CIM) compiler implemented with 3 nm high-κ metal gate (HKMG) FinFET technology, supporting flexible INT8 and FP16 formats for weight and activation multiply-accumulate (MAC) operations, offering configuration flexibility, high accuracy, and improved area and power efficiency. The FP16 digital computing in memory (DCIM) compiler achieves high arithmetic accuracy by adopting a multiplication-first (MF) FP computation flow, while the INT8 DCIM compiler provides improved power-performance-area (PPA). The DCIM compiler engine enables the automatic generation of flexible CIM macro configurations based on the tiling of block-level leaf cells. Our test chip in 3 nm technology is fully validated and exhibits 124.6 TOPS/W at 0.5 V and 90.2 TOPS/mm^2 at 1.1 V for INT8 DCIM, and 28.6 TFLOPS/W at 0.5 V and 17.4 TFLOPS/mm^2 at 1.1 V for FP16 DCIM, respectively.**

***Index Terms*—Digital-computing in memory (DCIM), DCIM compiler, floating-point (FP), integer, leaf-cell tiling, logic bitcell, multiplication-first, static random-access memory (SRAM).**

I. Introduction

As DNNs support increasingly complex image and language applications including generative AI, the accuracy, flexibility, and power efficiency requirements become more stringent. Compute-in-memory (CIM) has emerged as a promising solution, particularly for MAC-intensive computations. However, in analog-based CIM, realizing both better power-performance-area (PPA) and high accuracy is the major challenge for practical implementation. Recent works examined the trade-offs among multiply-accumulate (MAC)throughput, energy efficiency, and accuracy in analog-based or analog-digital mixed CIM designs [1], [2], [3]. On the other hand, digital computing in memory (DCIM) has gained attention for its ability to deliver massively parallel computation without any accuracy loss, leveraging distributed small static random-access memory (SRAM) arrays and customized MAC units as shown in Fig. 1. Fig. 2 illustrates the normalized TOPS/W and TOPS/mm^2 with technology scaling from 22 to 3 nm. An all-digital SRAM-based CIM macro was introduced at ISSCC 2021, and we have been exploring and developing DCIMs since 2020, with several papers published on the topic [4], [5], [6], [7], [8]. Over this period, DCIM achieved significant advancements in power, performance, and area efficiency, improving TOPS/W by approximately 5$\times$ and TOPS/mm^2 by over 10$\times$ as technology scaled from 22 to 3 nm node.

Although prior DCIM designs demonstrated excellent power efficiency, achieving a balance between hardware (HW) utilization and energy efficiency remains as one of the sig-

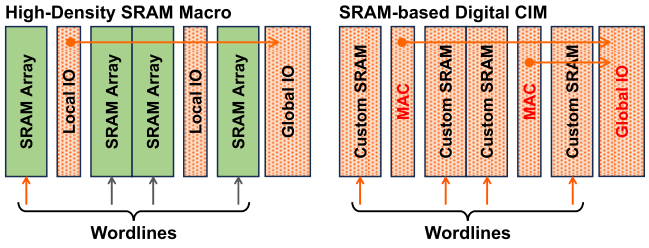


Fig. 1. Concept and floorplan comparison between high-density (HD) SRAM and SRAM-based DCIM.

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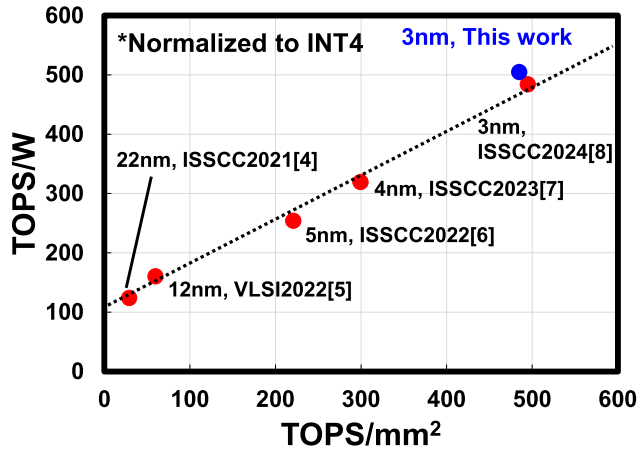


Fig. 2. Normalized TOPS/W and TOPS/mm^2 with technology scaling from 22 to 3 nm.

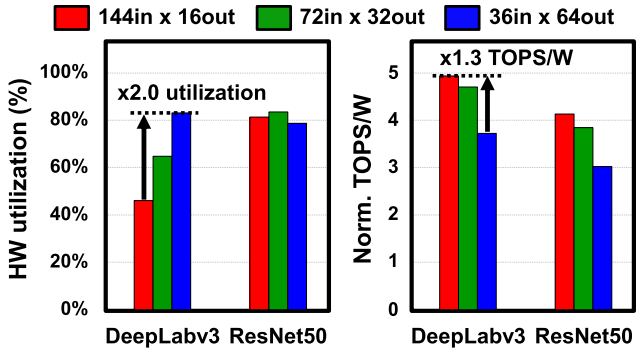


Fig. 3. Impact of macro dimensions on HW utilization and energy efficiency, TOPS/W.

nificant challenges. Fig. 3 presents our evaluation results using ML workloads [9]. The analysis reveals that smaller input channels tend to achieve higher HW utilization, while larger input channels enhance power efficiency. These trends also vary based on the type of neural network and the dimensions of the matrix operations involved. This highlights the need to design different macro sizes tailored to each network type, rather than relying on a fixed macro configuration.

Mobile devices are increasingly adopting a wide range of AI techniques, from low-precision tasks utilizing integer formats to high-precision operations using floating-point (FP) formats for weight training (e.g., face/image recognition, object detection, and language processing). Consequently, picking the most suitable bit precision for different ML applications is critical to achieve both higher arithmetic accuracy and better PPA at the same time. Table I summarizes the correlation between bit precision and neural network accuracy. FP16 allows AI models to operate without any loss in inference accuracy. On the other hand, using narrower bit widths significantly reduces memory usage and MAC complexity. For instance, INT8 achieves four to five times higher power efficiency (TOPS/W) compared to the FP16 format.

To address these challenges, we developed the DCIM compiler to offer the flexibility of picking most suitable

TABLE I
BIT PRECISION AND NETWORK ACCURACY

Inference Accuracy on ImageNet	MobileNet-v2	Inception-v1	EfficientNet-B1	MobileVit-v2	TOPS/W
FP32 SW Baseline	71.87	71.30	78.86	71.30	N/A
FP16 Accuracy	71.87	71.30	78.71	71.40	1x
INT16 Accuracy	71.69	71.29	78.70	71.40	~1.25x
INT8 Accuracy	69.90	70.99	78.01	66.50	~5x

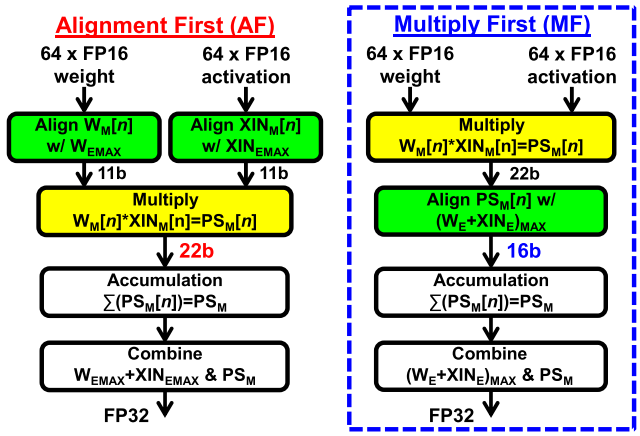


Fig. 4. Comparison of the two common flows of FP processing; the AF approach and the MF approach.

macro configuration. In this article, we present the first DCIM compiler that supports both INT8 and FP16 formats; INT8-optimized macros deliver excellent PPA, while FP16 macros ensure higher arithmetic accuracy. Our compiler engine automatically generates flexible CIM macro configurations by leveraging the tiling of the block-level leaf cells.

The remainder of this article is organized as follows. Section II discusses FP computation flows and presents our arithmetic accuracy analysis. Section III provides an overview of the architecture and operation of the INT8 and FP16 DCIM compiler. Chip implementation and Si measurement results are presented in Section IV. Section V summarizes the relevant findings of this work.

II. FLOATING POINT PROCESSING

A. Computation Flow

The complexity of FP computation primarily comes from the exponent alignment, prior to the accumulation. Multiplying two FP numbers involves two key steps: adding their exponent values and multiplying mantissa values. Before accumulation, the exponent values must be aligned by determining the maximum exponent value among all FP products. During alignment, the mantissa of each product is shifted by the exponent difference between its local exponent and the maximum exponent. These steps introduce significant complexity to the MAC implementation compared to the integer formats.

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Fig. 4 compares two common flows for FP computation. As shown in the left side, the alignment first (AF) approach aligns mantissa bits according to the maximum exponent (EMAX) values before multiplying the weights and activations (XIN). Applying the AF approach eliminates the need for shift operations related to exponent alignment, thereby allowing FP mantissa processing to utilize existing INT-based MAC units. Prior works demonstrated high-energy efficiency, TOPS/W, by employing weight and activation prealignment with off-macro digital circuits and software, which eliminates shift functions related to exponent alignment within their macro [10],[11], [12]. However, this still requires off-macro conversions for pre- and postalignment, which introduces additional overhead at off-macro, impacting the PPA. Furthermore, in terms of arithmetic accuracy, because the mantissa alignment and truncation are performed upfront, errors from both weight and XIN individually contribute to mantissa product. Accordingly, we observe a visible arithmetic accuracy loss with AF approach, which had been adopted by several prior designs [10], [11],[12], [13] using brain floating (BF) point format (featuring similar trends as FP format with respect to MF versus AF). In contrast, the multiply first (MF) approach performs mantissa multiplication before the bit alignment as shown in the right side. Therefore, it prevents information loss and preserves arithmetic accuracy for a 22 bit mantissa product (considering 11 bit mantissa for FP16, including the hidden bit information).

B. Arithmetic Accuracy

To evaluate the computation accuracy of the MF and AF approaches, we perform inference with each scheme across four representative ML models(i.e., MobileNet-v2, Inception-v1, EfficientNet-B1, and MobileViT-v2) and obtain the final outputs for each model. We use FP32-software as reference to compute the signal-to-quantization noise ratio (SQNR) [14] of model output accordingly, with following equation:

$$\mathbf{SQNR} = \mathbf{10log_{10}}\left(\frac{\mathbf{Output}_{\mathbf{FP32}}^{2}}{(\mathbf{Output}_{\mathbf{FP32}} - \mathbf{Output}_{\mathbf{CIM}})^{2}}\right). \quad (1)$$

Fig. 5 shows the SQNR-based arithmetic accuracy as a function of postaligned mantissa bitwidth. The results are averaged over the output SQNR of all four models, and each model is evaluated using the ImageNet dataset. The software baseline is running AI workloads inference with FP16 input and weight, whose products are in FP32 and will be further accumulated to FP32 output. For MF, we follow the exact HW scheme to run AI workloads and get accuracy. Compared with software baseline, the MF scheme with narrower bit loses information due to truncation of postaligned product mantissa and leads to accuracy drops. Considering the MF's SQNR is as high as the FP16 software for≥ 16 b (and visibly higher than the AF's SQNR with 22 b), in this work, we proceeded with MF flow featuring a mantissa bitwidth of 16 b before partial sum accumulation.

III. INT8 and FP16 DCIM Compilers

This section presents the architecture overviews and operations of INT8 and FP16 DCIM designs, along with the supported configuration ranges for the DCIM compiler. Table II summarizes the DCIM compiler features. In this design, two types of DCIM compilers are developed: the INT8 DCIM compiler and the FP16 DCIM compiler. Both INT8 and FP16 compilers share the same configuration ranges for input channels, output channels, and extra SRAM storage rows. Specifically, the input channel range is supported from 16 to 64 channels with a step size of 16 channels, the output channel ranges from 2 to 8 channels with a step size of 1 channel, and the additional SRAM storage rows from 2 to 18 rows with a step size of 2 rows.

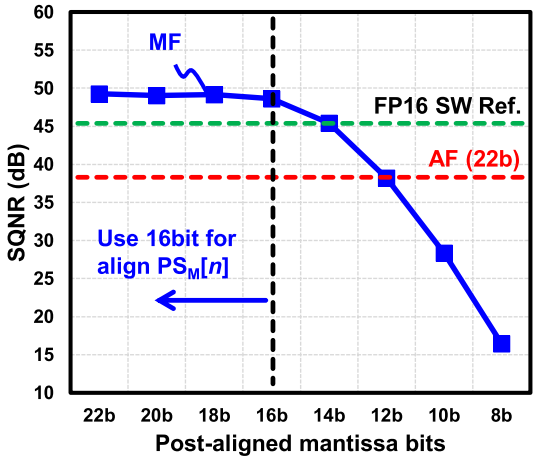


Fig. 5. Arithmetic accuracy based on the SQNR as a function of postaligned mantissa bitwidth.

TABLE II
DCIM Compiler Features

Compilers		INT8 DCIM	FP16 DCIM
Input format		Signed/unsigned INT8	FP16 & Signed/unsigned INT8
Output format		Signed/unsigned INT22	FP32 & Signed/unsigned INT22
Config.	# of input	16 ~ 64, w/ step16	
	# of output	2 ~ 8, w/ step1	
	# of rows	2 ~ 18, w/ step2	

A. INT8 DCIM Compiler

Fig. 6 presents an overview of the INT8 DCIM compiler in 3-nm technology. The largest DCIM macro configuration supports 64 input channels, 8 output channels, and 18 weight sets. Weight memory storage is partitioned into 32 banks. Each bank consists of 18 rows and 128 columns. A total of 128 columns include 8 bits for INT8, 2 input channels and 8 output channels (i.e., 8 × 2 × 8). The INT8 DCIM macro supports MAC operations with signed/unsigned INT8 for weight and XIN. The MAC unit supports parallel MAC scheme based on look-up-tables (LUTs) with a three-stage pipeline like the prior designs, comprising one SRAM array access stage and two MAC pipeline stages [8]. The MAC unit processes INT8 × INT8 parallel MAC with 4096 bits weight

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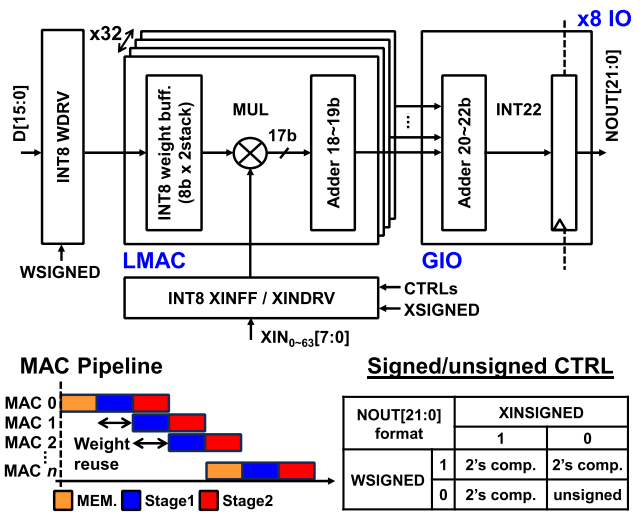


Fig. 6. Architecture overview of INT8 DCIM compiler design (top), Timing diagram of three-stage MAC pipeline and signed and unsigned format handling in INT8 DCIM (bottom).

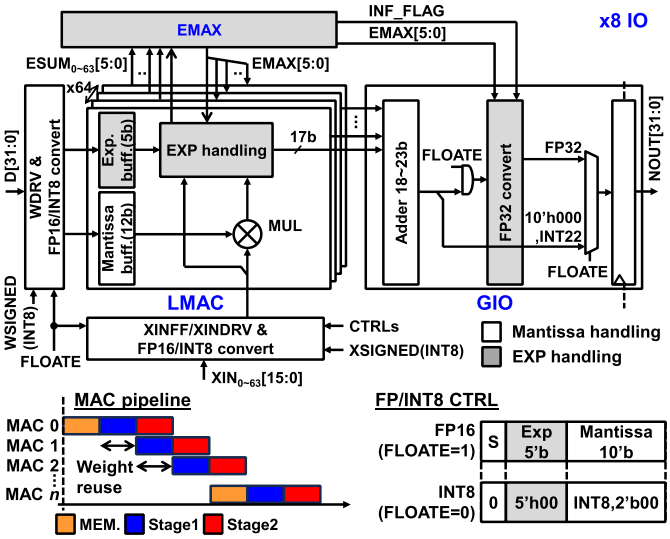


Fig. 7. Architecture overview of FP16/INT8 DCIM compiler design; exponent handling parts are newly adopted (top), timing diagram of three-stage MAC pipeline and FP16 and INT8 modes handling (bottom).

(8 bits × 64 input × 8 output channels) and 512 bits XIN (8 bits × 64 input channels). Each local-MAC (LMAC) includes 16 columns × 2 to 18 rows SRAM storage, 8 × 8 bit multiplier and two stage adder trees, and the global-IO (GIO)includes 20 to 22 bit sub-global and global adders. The signed/unsigned formats for both weights and inputs (XINs) can be switched independently using the control signals.

B. FP16 DCIM Compiler

As illustrated in Fig. 7, the FP16 DCIM supports FP16 and signed/unsigned INT8 formats for both weight and XIN. In this design, exponent handling with EXP addition/subtraction, EMAX search, mantissa shifter, and FP32 output format converters are newly adopted in the LMAC and GIO. The

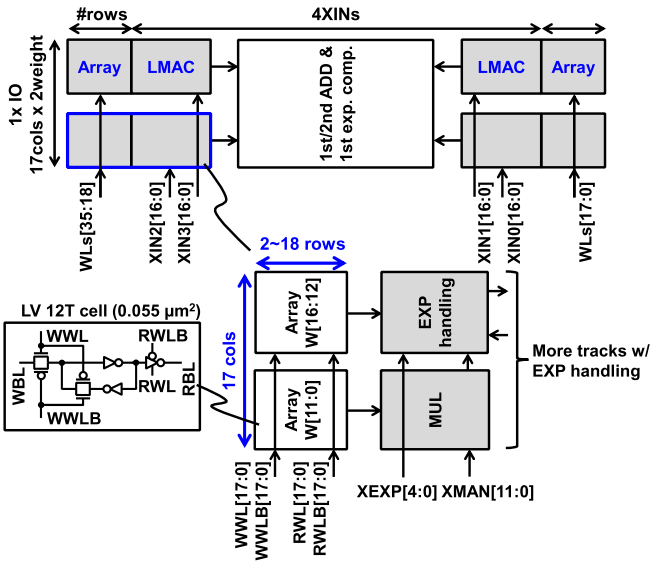


Fig. 8. Layout floorplan for the LMAC with two stacked weight storage and the LV 12 T bitcell schematic.

FP16 converter handles the hidden bit at the interface. The weight storage stores the 17 bits by appending one bit of hidden information. The FP16 × FP16 or INT8 × INT8 parallel MAC operations are handled by the MAC unit with 8704 bits weight (17 bits × 64 input × 8 output channels) and 1088 bits XIN (17 bits × 64 input channels). The FP computation flow is divided into two stages. In the first stage, multiplication is performed by adding two different exponent values and multiplying their corresponding mantissa values. The parallel MAC operation is implemented with a Booth multiplier to support mantissa alignment at each channel. The EMAX search module receives 384 bits (6-bit ESUM × 64 input channels) and sends the 6-bit EMAX to LMAC. An EXP subtraction block computes EXP difference, while a shifter block aligns and truncates mantissa to 16 bits based on this difference. After conversion to the 2's compliment format for accumulation, LMAC outputs a 17-bit product (1 bit-sign + 16 bit-data). The GIO includes 18 to 23 bit adders and FP32 output format converter. In the second stage, the adders accumulate partial sums (PSUMs) into 23 bits. The output format converter combines this 23-bit PSUM and 6-bit EMAX to produce a FP32 output (NOUT). The macro can operate in FP16 mode or INT8 mode, which is controlled by the FLOATE signal.

Fig. 8 illustrates the layout floorplan for the LMAC and bitcell array. We placed two weight data in vertical direction to mitigate routing congestion due to the adder tree and EMAX search. In order to support compiler function, the number of extra SRAM storage rows is made tunable from 2 to 18. In the largest macro configuration, weight memory storage is partitioned into 32 banks. Each bank consists of 18 rows and 272 columns. 272 columns include 17 bits for FP16, 2 input channels and 8 output channels (i.e., 17 × 2 × 8). A low-voltage 12 T bitcell [15] is adopted to enable simultaneous MAC + write functions, and wide range dynamic voltage-frequency scaling (DVFS) with V_{MIN} <0.5 V. The bitcell area is 0.055μm^2.

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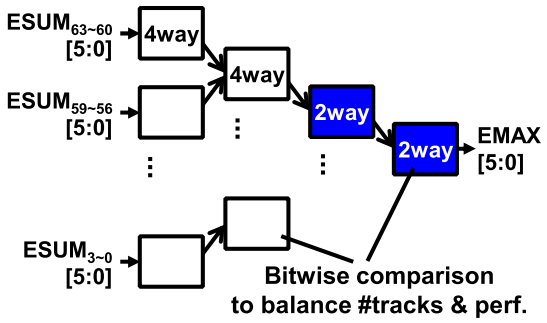


Fig. 9. Maximum exponent (EMAX) search with hierarchical and bitwise comparators.

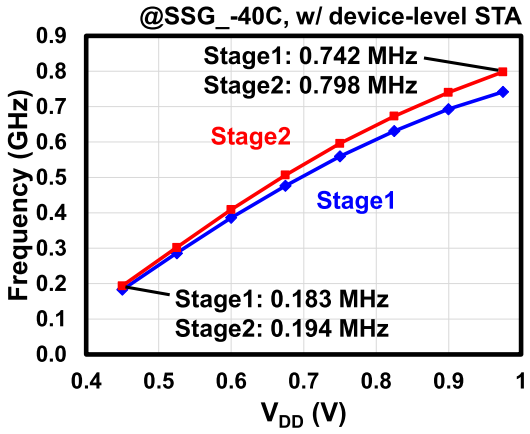


Fig. 10. Performance comparison between MAC stages 1 and 2 with STA.

Fig. 9 shows the block diagram of EMAX search with hierarchical and bitwise comparator. In the early stages of EXP search, a four-way comparator is used to minimize the number of gate delay stages for fast operation. On the other hand, bitwise comparison with a two-way comparator is adopted for the later stages of EXP search to minimize global wire tracks for better routing.

Fig. 10 illustrates the comparison of maximum operating frequency(f_{MAX}) between the first and second MAC pipeline stages across swept voltages. Device-level static timing analysis (STA) is used to evaluate critical path delays. We monitor the number of gate delay stages between MAC stage 1 (4-layer EMAX search + mantissa shifter) and the MAC stage 2 (adder tree + FP32 output format converter) are accordingly balanced. STA results confirm that the difference in the total number of gate delay stages is less than 5%, and the f_{MAX} of MAC stage 2 is within 7% of MAC stage 1 at 0.975 V.

C. Multi-Weight Update per Cycle

Fig. 11 (Top) shows the optimization of memory array. Running decoder-based generative AI models are highly memory bounded due to the lack of weight reuse. Prior designs [7], [8] updated weight data for one input channel per cycle (17 bits across 8 output channels). This limited bandwidth leads to computation stalls, which are particularly problematic for vector–matrix multiplications with no weight reuse. For

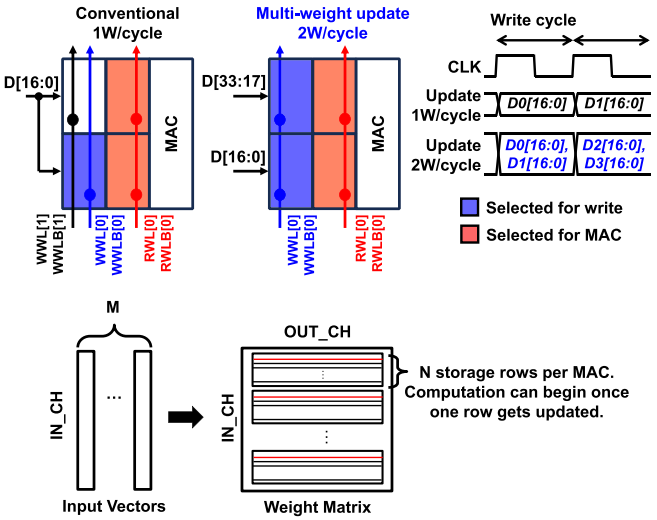


Fig. 11. Conceptual and timing diagram of the multi-weight update per cycle (top) and weight update data flow (bottom).

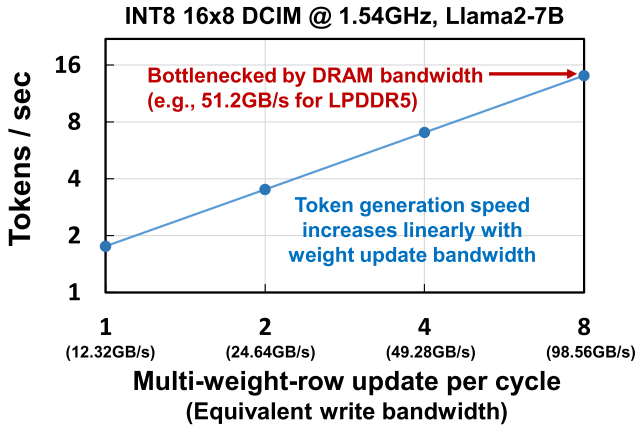
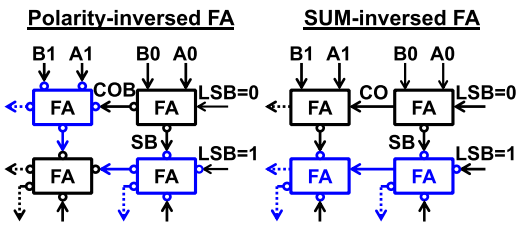


Fig. 12. Tokens/sec versus weight update BW for the decode phase of llama2-7B model.

instance, in an $M \times N$ DCIM macro, updating all weights takes M cycles, whereas computation requires just one cycle. To address this issue, we adopt a multi-weight update per cycle (e.g., 17 bits $\times$ 2 input channels across 8 output channels), doubling the write bandwidth and halving cycles, enhancing utilization and throughput with negligible area overhead.

Fig. 11 (Bottom) shows a weight update data flow. Our DCIM macro with multirow weight storage supports concurrent weight update and CIM. For a DCIM macro of size $IN_CH \times OUT_CH$ with storage row size of N, it can execute a matrix multiplication of size (M, IN_CH) $\times$(IN_CH, OUT_CH) in M cycles. Although updating all weights stored in N storage rows for all input channels takes $N \cdot IN_CH/2$ cycles (assuming 2 weight-row update per cycle), the computation can proceed once one weight set of size $IN_CH \times OUT_CH$ is updated (which takes$IN_CH/2$ cycles), while the subsequent set is being updated in parallel. Fig. 12 summarizes evaluation results of tokens/sec versus weight update bandwidth for the decode phase with llama 2-7B model. The throughput increases linearly with the weight

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Power & Perf. comparison

	Positive adder	Polarity Inversed adder	This work SUM inversed adder
Power	100%	94.0%	86.0%
Perf.	100%	87.7%	93.6%

Fig. 13. Power and performance comparison between proposed SUM-inversed FA and conventional polarity-inversed FA.

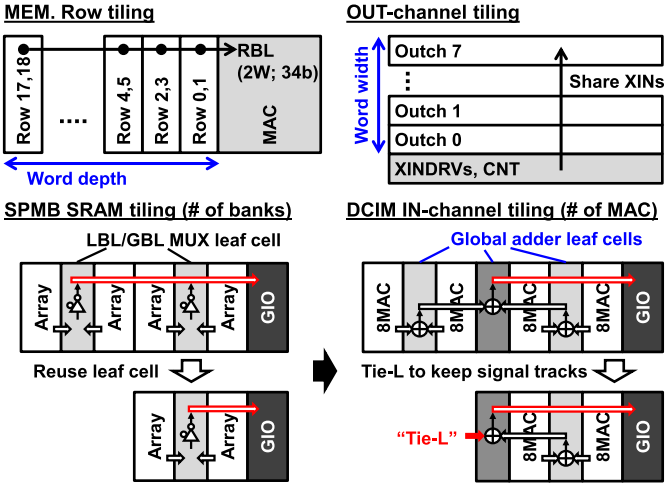


Fig. 14. Tiling method for DCIM compiler; the memory row tiling and the output channel tiling (top), and the input channel tiling (bottom).

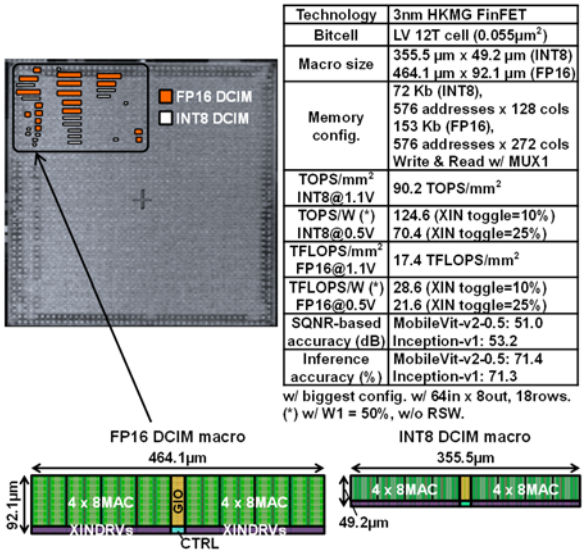
update bandwidth. For 8-weight updates per cycle, the write bandwidth achieves (INT8 $\times$ 8 out_ch $\times$ 8 sets $\times$ 1.54 GHz = 98.56 GB/s), which, however, will be bottlenecked by the DRAM bandwidth(e.g., 51.2 GB/s for LPDDR5). The area overhead is 5% in macro level (mainly from additional DFFs) in the case of 8-weight updates per cycle.

D. Circuit Optimization

As depicted in Fig. 13, a SUM-inversed full-adder (FA) is introduced to enhance low-power MAC operation. The SUM-inversed FA inverts only the SUM outputs while keeping a positive carry path, which ensures faster propagation along the carry direction. The MAC operation with the SUM-inversed FA achieves 8% improved power efficiency and 7% faster operation compared to the conventional polarity-inverted FA, which slows both carry and sum propagations due to the large fan-out, causing MSB side timing gap and glitches.

E. DCIM Tiling

Fig. 14 illustrates how tiling is supported in the DCIM compiler. Flexible DCIM macro configurations are automatically generated by the compiler engine based on block-level



Technology	3nm HKMG FinFET
Bitcell	LV 12T cell (0.055μm^2)
Macro size	355.5 μm x 49.2 μm (INT8) 464.1 μm x 92.1 μm (FP16)
Memory config.	72 Kb (INT8), 576 addresses x 128 cols 153 Kb (FP16), 576 addresses x 272 cols Write & Read w/ MUX1
TOPS/mm^2 INT8@1.1V	90.2 TOPS/mm^2
TOPS/W (*) INT8@0.5V	124.6 (XIN toggle=10%) 70.4 (XIN toggle=25%)
TFLOPS/mm^2 FP16@1.1V	17.4 TFLOPS/mm^2
TFLOPS/W (*) FP16@0.5V	28.6 (XIN toggle=10%) 21.6 (XIN toggle=25%)
SQNR-based accuracy (dB)	MobileVit-v2-0.5: 51.0 Inception-v1: 53.2
Inference accuracy (%)	MobileVit-v2-0.5: 71.4 Inception-v1: 71.3

w/ biggest config. w/ 64in x 8out, 18rows.
(*) w/ W1 = 50%, w/o RSW.

Fig. 15. Test chip micrograph alongside the layout plot and the specification table.

leaf cells. In order to implement leaf cell tiling, we defined three configuration parameters. The number of rows and output channels correspond to the word depth and word width in the SRAM compilers, respectively. The number of input channels can be handled as similar to the number of banks in single-port multibank (SPMB) SRAM design. In the SPMB, reuse of leaf cells is convenient because global routing can be shared among the leaf cells in configuration range. On the other hand, in the DCIM, global adders need extra tie-low leaf cells to maintain global routing since the number of input channels is reduced. For instance, when the number of input channels decreases from 64 to 32, the final-stage adder requires tie-low leaf cells to maintain the global routing, it incurs slight overhead on the PPA. In addition to the tie-low cells, peripheral logic (e.g., GIO, WL/XIN DRVs, GCTRL) involves additional overheads especially for the smaller output channel configurations. We set the compiler range for the extra SRAM storage rows from 2 to 18 with a step size of 2 rows, the number of output channels from 2 to 8 with a step size of 1 channel, and the number of input channels from 16 to 64 with a step size of 16 channels.

IV. Chip Implementation and Measurement Results

Fig. 15 presents a die photograph of the test chip for the FP16 and INT8 DCIM compiler implemented in 3 nm FinFET technology, along with a specification table. Macro area for the largest configuration of FP16 DCIM compiler is 464.1 $\times$ 92.1 μm (=0.04274 mm^2), and macro area for the largest configuration of INT8 DCIM compiler is 355.5 $\times$49.2 μm (=0.01749 mm^2). The maximum bit capacity per macro is 153 kb for FP16 DCIM compiler and 72 kb for INT8 DCIM compiler.

Since BIST and DFT schemes are key components for mass production, we implemented CIM BIST based on a commercial MBIST engine. The CIM BIST includes a custom algorithm for the DCIM macro and realizes > 99%

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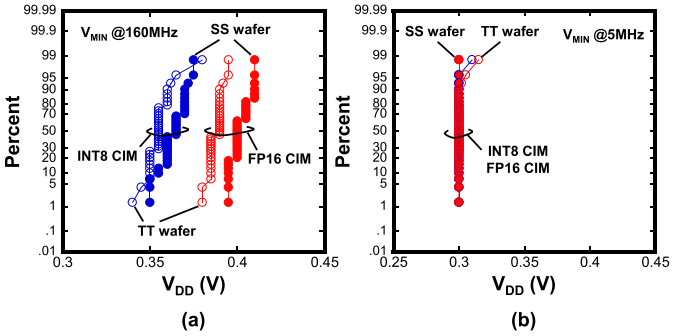


Fig. 16. Measured V_{MIN}. (a) V_{MIN} at testing frequency of 160 MHz and (b) V_{MIN} at testing frequency of 5 MHz.

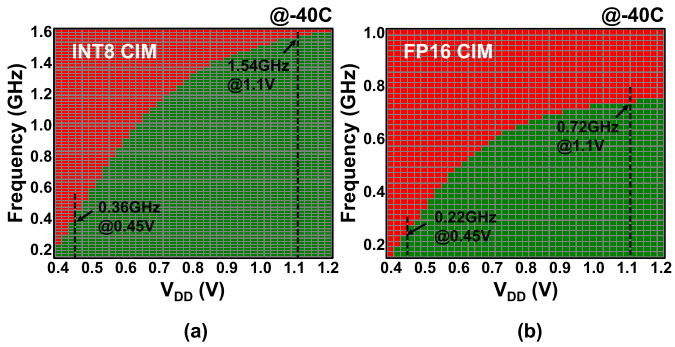


Fig. 17. Measured shmoo plot and f_{MAX}. (a) f_{MAX} for INT8 DCIM and (b) f_{MAX} for FP16 DCIM.

fault coverage in gate-level fault simulation. The CIM BIST can achieve faster silicon feedback after the fabrication. On the other hand, the MBIST + ATPG testing method has better debug capability with custom pattern generation. We also utilize MBIST for memory and the ATPG for MAC logic. The CIM BIST can verify each macro with testing time < 2 ms.

Fig. 16(a) and (b) summarizes the minimum operating voltage(V_{MIN}). We observed that V_{MIN} is 0.37 and 0.41 V for INT8 and FP16 DCIM, respectively, at 160 MHz and −40 °C with 95% yield. The V_{MIN} for FP16 is limited by CIM f_{MAX} during measurements at a testing frequency of 160 MHz. When the testing frequency is slowed down to 5 MHz, both INT8 and FP16 DCIM achieve an equal V_{MIN} of 0.30 V at −40 °C with 95% yield. Fig. 17(a) and (b)shows measured shmoo plot and f_{MAX}. We confirmed f_{MAX} with 1.54 and 0.72 GHz for INT8 and FP16 DCIM, respectively, at 1.1 V and −40 °C. We observed that the f_{MAX} of INT8 MAC mode in the FP16 DCIM macro is 0.84 GHz, at 1.1 V and −40 °C (=16.6% faster than the FP16 MAC mode).

Fig. 18(a)–(d) presents the power efficiency (TOPS/W versus voltage)for INT8 DCIM and FP16 DCIM. The weight activation ratio is 50%and different toggle rate, 10% and 25%, for XIN are considered. The TOPS/W increases when the toggle rate and/or voltage is low. A toggle rate with 10% results in 124.6 TOPS/W with INT8 DCIM and 28.6 TFLOPS/W with FP16 DCIM at 0.5 V and no memory access condition. We observed a good correlation between the

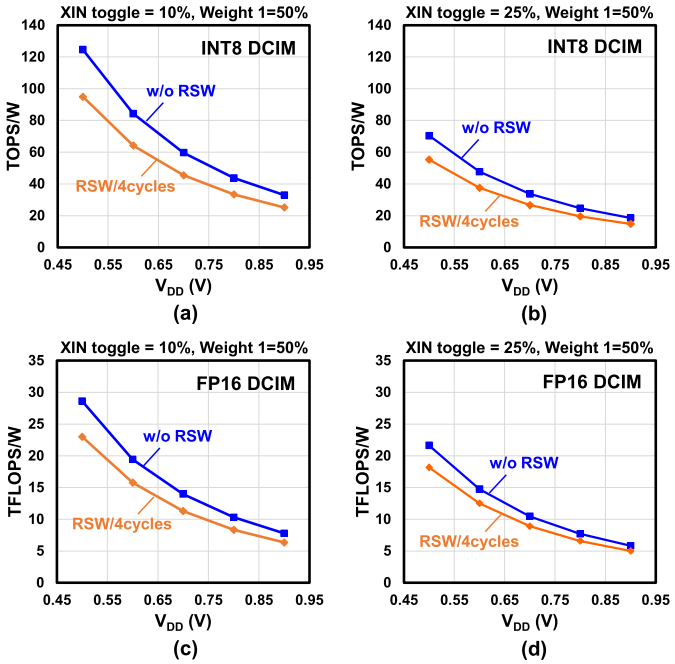


Fig. 18. Power comparison with voltage sweep, different XIN toggle rate, and different memory row switch frequency (RSW). (a) and (b) INT8 DCIM with XIN toggle = 10% and 25% and (c) and (d) FP16 DCIM with XIN toggle = 10% and 25%.

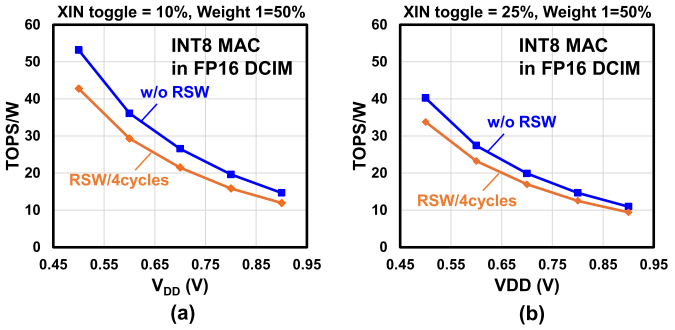


Fig. 19. Power comparison for INT8 MAC mode in FP16 DCIM. (a) XIN toggle = 10% and (b) XIN toggle = 25%.

simulated and measured power. Fig. 19 presents the power efficiency with the INT8 MAC mode in the FP16 DCIM macro. The power efficiency is improved by 1.86× compared to the FP16 MAC mode. We observed that a toggle rate with 10% results in 53.2 TOPS/W with INT8 MAC in the FP16 DCIM at 0.5 V and no memory access condition.

Fig. 20 presents PPA with different macro configurations. Switching from 18 to 2 rows configuration improves area efficiency, while reducing the macro's memory capacity. Area and power overhead from peripheral logic (e.g., GIO, WL/XIN DRVs, GCTRL) are increased when the macro configuration is reduced for the input channel from 64 to 16 and the output channel from 8 to 2, compared to larger macro configurations (64 in_ch × 8 out_ch).

Table III compares this work with prior art. The DCIM macro with LV 12 T bitcell enables ultralow voltage operation and wider DVFS from 0.3 to 1.1 V. The DCIM compiler

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TABLE III
PPA SUMMARY TABLE AND COMPARISON TO PRIOR WORKS

		ISSCC'24 [8]	ISSCC'23 [10]	ISSCC'23 [11]	This work
Technology		3nm	22nm	28nm	3nm
Bitcell		6T	6T	6T + FCU	LV 12T cell
VDD		0.36 ~ 1.1V	0.6 ~ 0.8V	0.6 ~ 0.9V	0.3 ~ 1.1V
Compiler support		No	No	No	Yes
Input Ch range		72	128	128	64 ~ 16ch, step16
Output Ch range		4	32	8	8 ~ 2ch, step1
#rows range		18	8	8	18 ~ 2 rows, step2
INT	Macro area	0.0167 mm^2	N/A	0.146 mm^2	0.0175 mm^2 (*)
	Input format (W/XIN)	INT12/INT12		INT8/INT8 BF16/BF16	INT8/INT8
	Output format	INT31		INT23 or BF16	INT22
	TOPS/W	32.5 (XIN toggle 10%) @0.55V		44 (XIN1=50%) @0.6V	@0.5V (*) 124.6 (XIN toggle=10%) 70.4 (XIN toggle=25%)
	TOPS/mm^2	55.0 @0.9V		2.5 @0.9V	90.2 @1.1V (*)
FP	EXP alignment	N/A	Alignment first	Alignment first	Multiply first
	Macro area		N/A	0.146 mm^2	0.0427 mm^2 (*)
	Input format (W/XIN)		BF16/BF16	INT8/INT8 BF16/BF16	INT8/INT8 FP16/FP16
	Output format		FP32	INT23 or BF16	INT22 or FP32
	TFLOPS/W		@0.6V, BF16 70.21 (XIN1=10%)	@0.6V, BF16 31.6 (XIN1=50%)	@0.5V (*) 28.6 (XIN toggle=10%), FP16 21.6 (XIN toggle=25%), FP16 53.2 (XIN toggle=10%), INT8 40.2 (XIN toggle=25%), INT8
	TFLOPS/mm^2		N/A	2.0 @0.9V, BF16	17.4 @1.1V, FP16 (*) 20.3 @1.1V, INT8 (*)

(*) with biggest macro configuration (64in x 8out, 18rows).

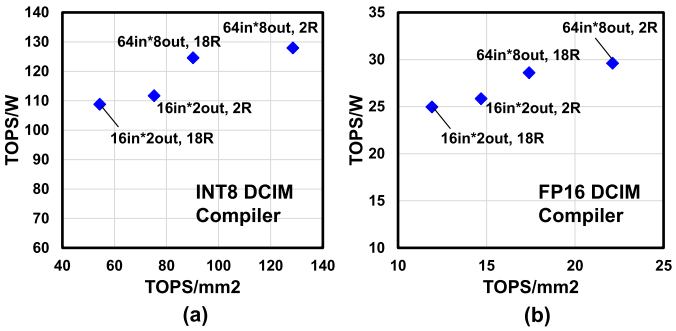


Fig. 20. PPA with different configurations. (a) INT8 DCIM compiler and (b) FP16 DCIM compiler.

supports flexible configuration ranges for the input and output channels, and the number of additional SRAM rows. The prior works employed the AF approach and eliminates the need for shift operations related to exponent alignment, have demonstrated high-energy efficiency, TOPS/W as their weight and activation are prealigned at off-macro digital circuits and software. However, this still requires off-macro conversions for pre- and postalignment which introduces additional overhead at off-macro. Our FP16 DCIM compiler demonstrated higher arithmetic accuracy with the MF FP computation flow, which incorporates a 16-bit mantissa bit width, while the INT8 DCIM achieves comparable PPA to the rigid INT8 DCIM design in [8].

V. CONCLUSION

We presented the first DCIM compiler offering flexible configuration range and high-compute precision, and improved PPA in 3 nm technology. Our INT8 DCIM offers better energy efficiency and the FP16 DCIM compiler offers higher arithmetic precision by adopting the multiply first approach. Our compiler engine automatically generates flexible macro configurations. Our 3 nm test chip is fully validated by using CIM-BIST/ATPG, and demonstrated 124.6 TOPS/W at 0.5 V and 90.2 TOPS/mm^2 at 1.1 V for INT8 DCIM, and 28.6 TFLOPS/W at 0.5 V and 17.4 TFLOPS/mm^2 at 1.1 V for FP16 DCIM, respectively.

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